RV College of Engineering®
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1RV22CS067

Academic year 2025-2026 (ODD Sem)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Date: 15/12/2025	**CIE - II**	**Max. Marks:(10+50) = 60**
Semester: VII	**UG**	**Duration: 2 Hours**
Course Title: Computer Graphics & Virtual Reality		**Course Code: CS373IA**

Q. No.	PART-A	M	BT	CO
1.	A line joining a point P and Q has its equivalent parametric equation $x(t)=2+6t$ and $y(t)=10-8t$ where $0\leq t\leq 1$. What are the coordinates of P and Q.	1	2	1
2.	Implicit form of the line joining two points P and Q is given by $6x-10y+60=0$. What is the equivalent parametric equation of this line for the parameter $0\leq t\leq 1$.	1	2	2
3.	Consider the scan conversion of a line whose slope is greater than -1 and less than 0 by Bresenham's line drawing algorithm. If (x, y) is the current pixel plotted and p is decision parameter. If $p \geq 0$ then what will be the p value for next iteration.	1	1	1
4.	Consider the midpoint circle generating algorithm which scan converts the circle octant from 0 to 45^0. If p is the decision parameter and (x, y) is the previous pixel plotted. If $p<0$ then what is the next pixel to be plotted.	1	2	1
5.	If we want to recolor an area that is not defined within a single-color boundary, then which filling algorithm to be used?	1	1	1
6.	A clipping window has bottom left corner at (3, 4) and top right corner at (10, 9). Using Cohen-Sutherland line clipping algorithm find what are the region codes of the end points of the line from A(2, 11) to B(9, 2).	1	3	2
7.	Liang-Barsky algorithm is used to clip a line from A(-15, -30) to B(30, 60) against the window having diagonally opposite corners as (0, 0) and (15, 15). What are the coordinates of the visible portion of the clipped line?	1	3	2
8.	The triangle [ABC] with the coordinates A(5, 4), B(8, 3) and C(8, 8). Rotate this triangle by 180^0 about its centroid. What are the coordinates of the rotated triangle?	1	4	3
9.	The square [ABCD]=[(2, 2), (4, 2), (4, 4), (2, 4)] is rotated by 90^0 will have the transformed coordinates [A'B'C'D']=[(-2, 2), (-4, 2), (-4, 4), (-2, 4)]. What are the coordinates of the square if it is reflected about the axis y=-x.	1	3	2
10.	The triangle [ABC]=[(0, 0), (2, 2), (10, 4)] is required to be magnified to four times of its original size while keeping C(10, 4) fixed. What are the coordinates of the triangle after magnification?	1	2	2

Q. No.	PART-B	M	BT	CO
1.	Explain Bresenham's Line drawing algorithm by showing all the deviations for the decision parameters, to scan convert a line whose slope m is $0 \leq m \leq 1$. How to modify this algorithm for the lines of other slopes. Scan convert a line from (10, 25) to (20, 12) on a raster screen using Bresenham's line drawing algorithm. Compare it with line generated using DDA algorithm.	10	4	4
2a.	Explain Mid-Point Circle drawing algorithm by showing all the deviations for the decision parameters to scan convert a circle whose center is at (x_c, y_c) and a radius r. Plot circle centered at (5, 5) having r=8.	5	4	4
2b.	Explain the Scan-Line Polygon Fill Algorithm for any given polygon. List the benefits and drawbacks of this algorithm.	5	3	2
3	Explain Liang-Barsky line clipping algorithm to clip a line w.r.t an rectangular window boundary. Give the algorithm (C-Code) for the same. Let R be the rectangular window whose lower left hand corner is at L (1, 2) and upper right hand corner is at R (9,8). Use Liang-Barsky algorithm to clip the following line segments. Line 1 from A(-1,7) to B(11,1), Line 2 from C(3,7) to D(3,10) Line 3 from E(2,3) to F(8,4), Line 4 from G(11,6) to H(11,10)	10	4	3
4a.	Determine the homogeneous transformation matrix for reflection about the line y=mx+c. Reflect the triangle ABC[(2,3), (5,8), (7,2)] about a) The horizontal line y=2. b) The vertical line x=2. c) The line y=x+2. d) Draw the reflected image in all the cases.	6	2	2
4b.	Consider a line AB[(3, 2), (8, 7)] in right handed coordinate system. Rotate this line such that A lies on the origin and B lies on positive y axis. Find the transformation matrix and the resulting line.	4	4	3
5a.	Apply a suitable 3D transformation to rotate a pyramid defined by the coordinates A(0, 0, 0), B(1, 0, 0), C(0, 1, 0) and D(0, 0, 1) by 45 degrees about the line P(1, 1, 1) to Q(2, 3, 4).	6	2	2
5b.	Devise a method to reflect a 3D object about an arbitrary plane.	4	3	2

Course Outcomes: After completing the course, the students will be able to: -	
CO1:	Apply the basic concepts of Computer Graphics & Virtual Reality which illustrates the use of the pipel architecture, OpenGL library, VR tools.
CO2:	Analyze and make an appropriate choice of methods required for computer representation of 2D/3D obj and modelling in Virtual Reality.
CO3:	Design Computer Graphics and Virtual Reality applications using OpenGL library & VR tools.
CO4:	Implement common geometric construction & VR techniques and develop Computer Graphics/ VR b solution to Engineering applications / real-world problems.
CO5:	Exhibit team work and effective oral/written communication skills to accomplish a common goal of solv Computer Graphics/Virtual Reality problems with the engineering community and society at large.